

An Analysis of Information Types and Cognitive Activities Involved in Cross-session Search

Yuan Li, Austin R. Ward, Rob Capra

University of North Carolina at Chapel Hill

yuanli@email.unc.edu, austinrw@unc.edu, rcapra@unc.edu

ABSTRACT

Previous studies found that cross-session search tasks are often complex due to the uncertainty of task features, users' perception of the tasks, and the challenges (e.g., time limitation, interruptions) that users encounter during the process. Using Anderson & Krathwohl's taxonomy of learning [2], we analyzed the complexity of real-life cross-session search tasks reported by 110 Amazon Mechanical Turk workers. We found that cross-session search tasks often require different types of information (i.e., factual and conceptual information, procedure information, opinions, and information for helping metacognition) and are frequently associated with higher-level cognitive activities, including: applying analyzing, evaluating, and creating. Our findings add knowledge about cross-session search, and have insights for future research related to system design and studying cross-session search.

KEYWORDS

Cross-session search; Information type; Cognitive activity; Task complexity

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1 INTRODUCTION

Cross-session searches (XSS) are a common phenomenon in daily life when people conduct a series of information search activities across multiple sessions (e.g., time periods) to achieve a single goal [21]. Cross-session search is often associated with *complex* tasks [1, 7, 14]. While previous studies have revealed that XSS tasks cover a variety of topics and goals (e.g., gathering information to plan a trip, working on a long-term research project, or collaborating with others [5, 16, 20, 21]), relatively less work has focused on how specific cognitive activities and types of information are involved in XSS. That is the main focus of this paper.

To better understand the complexity of XSS tasks (and to develop effective support tools), it is important to consider the sources

that introduce the complexity of XSS tasks. Task complexity has been studied from different perspectives, including objective task features, users' capabilities, and their perceptions of difficulties when conducting the tasks [3, 4, 15]. Task complexity can also be studied through the lens of learning objectives: the types of information/knowledge involved and the required cognitive activities to complete the task. Researchers have found that complex tasks often involve gathering different types of information and require higher-level of cognitive process to complete [8, 9, 23].

To investigate the complexity of XSS, we analyzed 110 real-life XSS tasks reported by Amazon Mechanical Turk workers in a recent survey study [11]. We address the following two research questions: **RQ1**: What types of information do users need and search for in their everyday cross-session search tasks? **RQ2**: What types of cognitive activities do users perform during cross-session searches?

2 RELATED WORK

2.1 Cross-session search

Cross-session search describes situations when people conduct a series of information search activities across multiple different sessions to achieve a single goal [21]. Sellen et al. [20] found that 40% of knowledge workers' information-gathering tasks were spread over time. Morris et al. [18] reported that 83% of surveyed information workers had engaged in XSS. Recent research found that users often switch between multiple tasks and devices to complete searches across multiple sessions [25, 26].

Previous research found that XSS often involves *complex* tasks. Researchers have investigated the characteristics of XSS tasks in terms of topics, search sessions, time gaps, search behaviors, and reasons that lead people to search across sessions (e.g., [1, 13, 14, 17, 22]). Studies found that information gathering tasks often cause XSS because they require searchers to collect information from various sources [16, 20]. Research has found XSS may involve anywhere from 2 to 100 search sessions [21], but results from a diary study found that on average, users required from 2 to 3 sessions to complete their tasks [16]. Similarly, the time gap between sessions may vary, but a survey study found that the majority (75%) of XSS time gaps fall between hours to days [18].

Prior research has also explored the reasons that cause XSS. Spink et al. [22] found that academic searchers often resumed searches to enhance or refine what they saw earlier or to find additional information. MacKay and Watters [17] observed that students often stopped XSS because they found what they needed, or because the tasks were completed. Lin [12] developed the Multiple Information Seeking Episode (MISE) model and conceptualized eight task-related resuming reasons, such as: task spawns sub-problems, previously found information does not work, or people need updated information, etc. Lin [12] also distinguished two types of

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reasons that cause people to stop during XSS, including external reasons like interruptions, and internal reasons initiated by the searcher themselves. A recent study confirmed that many of the reasons mentioned above are observed in real-life XSS examples [11].

2.2 Task complexity

Task complexity is an important factor that can affect information search behavior and task performance [3]. Campbell [4] defined task complexity as the uncertainty of task outcomes, the paths/methods to achieve the desired results, and the relations (e.g., conflicting, interdependent) between the outcomes and the paths. Byström and Järvelin [3] developed the concept of *a priori determinability* to refer the degree of uncertainty about task inputs, process, and outcome. They also identified three types of information used by professionals: problem information (PI), domain information (DI), and problem-solving information (PSI). They found that complex tasks are often associated with higher levels of uncertainty due to unstructured task requirements and processes [3].

Jansen et al. [8] and later Kelly et al. [10] adapted the *cognitive complexity* taxonomy from Anderson and Krathwohl's [2] learning objectives to investigate the effects of task complexity on users' search behaviors. Anderson and Krathwohl's taxonomy includes cognitive complexity levels (from lowest to highest level of complexity): remember, understand, apply, analyze, evaluate, and create. Researchers have found that tasks with a higher-level of cognitive complexity could trigger more search activities [8, 10].

Recent research [24] used both the *cognitive complexity* and *knowledge type* dimensions of Anderson and Krathwohl's taxonomy to design search tasks and found that knowledge types (i.e., factual, conceptual, procedural, metacognitive) had significant effects on participants' perceptions of task difficulty, their performance, and search behaviors. Choi et al. [6] classified four types of information from objective to more subjective (i.e., facts, concepts, opinions, and insights) and found that more cognitively complex tasks require a greater variety of information types and more subjective information than the less complex tasks.

While the prior research efforts provide insights about XSS and a lens for studying task complexity, less attention has been given to understanding the complexity of XSS, particularly from the information types and cognitive activities involved in XSS.

3 METHOD

Survey design and data collection— We designed a survey study using the Critical Incident Technique (CIT) intended to identify important aspects of the XSS people experienced in their daily work and life. The questionnaire¹ contained four sections: (1) demographic information (i.e., age, gender, education background); (2) questions about a recent task or project that involved searching for information across multiple sessions (e.g., task description, task goal, and the information needed to complete the task); (3) questions about the *Most Recent Search Session* (MRS) the participant conducted for the selected task (e.g., time when the MRS happened, information they searched, results of the search session,

reasons they started and stopped the MRS); (4) information keeping methods used through XSS. Our question formats included open-response questions, multiple-choice questions, and 7-point Likert-type questions. We distributed the survey on Amazon Mechanical Turk (AMT) across 14 days in multiple batches at different times of day to attract workers from diverse groups.

Data analysis— We analyzed each participant's response to four open-ended questions: (1) their description of the work task; (2) their work task goals; (3) the needed information for completing the task; (4) the description of the MRS. Specifically, we focused on participants' descriptions about the information they wanted/needed, what they found online, and how they used the found information.

We coded the *cognitive activities* and *types of information* mentioned across users' survey responses. Following Urgo et al. [23] and Choi et al.'s [6] work, we adapted Anderson and Krathwohl's [2] learning objectives (i.e., remember, understanding, apply, analyze, evaluate, create) as our codes for analyzing the cognitive complexity of XSS tasks. For coding the *types of information*, we created a set of codes based on Anderson and Krathwohl's [2] classification of the types of knowledge (i.e., factual, conceptual, procedural, metacognitive), combined with the information types distinguished by [6] (i.e., fact, concept, opinion, insights). Our final set of codes for classifying the information types included: factual and conceptual, procedural, metacognitive, and opinion. We define all our coding definitions in Section 4. One of the authors coded the responses of all 110 participants, and another author verified the coding results independently. Then the two coders met and discussed the codes until complete agreement was achieved through consensus.

4 RESULTS

We received 110 complete responses that qualified for the final analysis. Overall, the response quality was high. Eighty percent (80%) of participants spent more than 10 minutes on the survey. Responses averaged 210 words in total for the four open-ended questions. Participants reported as 45% female, an average age of 35 (SD=10), a wide range of educational levels (from high school to graduate level), and more than 50 different occupations [11].

4.1 Information types involved in XSS

4.1.1 Factual and conceptual information. Factual and conceptual information refers to information elements and the relationships among factual elements. Table 1 shows that factual and conceptual information is the most popular involved information type during XSS: 96% (N=106) of our participants searched for factual and conceptual information. The factual information participants searched includes names of specific things (e.g., people, plants, computer parts), locations, prices, policies, illness, etc. Conceptual information includes descriptions of relationships, such as the growth phases of plants (for gardening), causes of a particular illness, and distinctions between concepts, etc.

4.1.2 Procedural information. Procedural information describes the processes of how something happens or methods to complete a task. As the second most popular information type involved during XSS, 43% (N=47) participants mentioned searching for procedural information. For instance, participants looked for procedural information about how to build a computer, how to grow a successful

¹https://ils.unc.edu/searchstructures/resources/li_chiir2021_xss_questionnaire.pdf

garden, how a doctor conducts a knee surgery (because the person him/herself was going to have the surgery), etc.

4.1.3 Opinion. Opinion information was coded as reviews, perspectives, and judgements shared by others. Opinions can range from objective (e.g., official reports, academic papers) to more subjective ones (e.g., personal experience or feelings). Forty-two percent (N=46, 42%) of our participants mentioned that they searched for opinions for their XSS tasks. Examples included: book reviews, discussions posted on blogs/forums, professional recommendations for medical issues, evaluations of different products, and insights/tips about completing processes.

Table 1: Information types involved in XSS

Information Types	Participants	Percentage
Factual and conceptual	106	96%
Procedural	47	43%
Opinion	46	42%
Metacognitive	13	12%

4.1.4 Information for helping metacognition. Information for helping metacognition refers to information that people looked for to help them understand information that they had found. This could include different ways of presenting a concept, or documents that presented information that someone else had already aggregated or synthesized. In many of these cases, the searcher recognized that they wanted extra information to assist the interpreting and understanding process (e.g., through metacognitive monitoring or reflecting). Some participants (N=13, 12%) explicitly reported searching for information that can assist their cognitive activities (e.g., understand, analyze, evaluate). For instance, one participant described searching for videos after she found a recipe because she felt the videos would provide more visual help for understanding than the text (p32); another participant looked for a page that compared the pros and cons of computer parts although he had already collected a lot of information about different products (p12).

Table 2: Number of different information types involved in XSS

# Information Types	Participants	Percentage
Involved Info. Types =1	36	33%
Involved Info. Types =2	50	45%
Involved Info. Types =3	20	18%
Involved Info. Types =4	4	4%

4.1.5 Multiple Information Types Needed. We also investigated the number of different information types a participant needed/ searched during their XSS process. Table 2 shows that more than half (N=74, 64%) of participants mentioned that they needed at least two types of information for their XSS tasks. We found that participants who mentioned only one type of information (N=36, 33%) were usually looking for many different pieces of information with the same type, especially for factual and conceptual information. For instance, a person who wanted to buy a car insurance searched for different companies, their policy quotes, the coverage and offer types, and documents needed for applying the insurance.

4.2 Cognitive activities involved in XSS

4.2.1 Remember. In previous research that has used Anderson & Krathwohl's taxonomy applied to interactive information retrieval,

the *remember* level of cognitive complexity has been adapted for describing tasks with the least cognitive complexity level (e.g. finding specific facts online [9]). Following this definition, we found that 73% (N=80) of participants' XSS tasks involved finding the factual/conceptual information they needed (Table 3). We did not find remember-related actions mentioned by the other 28 participants because: (1) they had not found the needed factual/conceptual information yet, or (2) they found something relevant but still wanted to search for more.

Table 3: Types of cognitive activities involved in XSS

Cognitive Activity Types	Participants	Percentage
Remember	80	73%
Understand	38	35%
Apply	53	48%
Analyze	43	39%
Evaluate	71	65%
Create	38	35%

4.2.2 Understand. *Understanding* addresses the sense-making behavior when searchers interpret, exemplify, or infer the information they found. Thirty-five percent (N=38) of participants clearly stated how they constructed meaning based on the retrieved information to learn about the task topic and understand how different parts function together. We speculate that this percentage may be low since *understanding* is an integral part of many complex search tasks, participants did not explicitly mention it.

4.2.3 Apply. *Apply* describes when the searcher used the found information to carry out procedures, make decisions, and solve problems. We note that our threshold for classifying a description as "apply" required participants to mention utilizing the information in specific situations. In other words, we looked for evidence that participants had used the information to achieve concrete goals. Forty-eight percent (N=53) of participants reported that, as part of their XSS tasks, they applied the found information to accomplish specific tasks such as making purchase decisions, sharing the information with clients, following steps/tips to complete a job.

4.2.4 Analyze. *Analyze* refers to mental activities like breaking down information into parts or examining different aspects related to the task. Thirty-nine percent (N=43) of participants described analyzing activities as part of their XSS tasks. When working on tasks with multiple requirements, people often described breaking the requirements into multiple components. Sometimes they needed to compare the found information to select the most appropriate parts and organize them into a coherent piece. For instance, a person who was planning a trip searched across multiple sessions for locations, lodging, attractions, and other information to "bring them all together and create an itinerary" (p04).

4.2.5 Evaluate. *Evaluate* happens when people make judgments based on criteria or standards by comparing, monitoring, and critiquing [2]. Sixty-five percent (N=71) of participants' responses included descriptions of evaluating activities, making it the second most popular cognitive activity in XSS. This might be because XSS tasks often require evaluating information from different sources. As people encounter similar information, they need to critique and make judgements. For instance, a person looking for a job evaluated different openings based on pay, positions, locations with the

goal to “find a job that was close to me, pays well, and have good reviews” (p03).

4.2.6 Create. As the most complex cognitive activity, *create* is about using one’s own knowledge and skills to put the found information together to form something new. During the process of creating, it is important to utilize one’s own intelligence and make adjustments based on specific task situations. Thirty-five percent (N=38) of our participants described creating something new using the information found online, such as writing a book report or a research proposal, making a plan for a vacation, building a customized computer, or setting up a saltwater aquarium.

Table 4: Number of cognitive activity types involved in XSS

# Cognitive Activity Types	Participants	Percentage
Involved Cog. Types =1	11	10%
Involved Cog. Types =2	29	26%
Involved Cog. Types =3	36	33%
Involved Cog. Types =4	23	21%
Involved Cog. Types =5	10	9%
Involved Cog. Types =6	1	1%

4.2.7 Multiple Cognitive Activities. We also analyzed the number of different cognitive activities involved for each participants’ XSS process. Table 4 shows that the majority of participants described between 2 to 4 different types of cognitive activities involved during their XSS. For instance, a participant who was writing an article about the history of the United States Space Shuttle program searched for information about the astronauts, their families, images, and official reports (*remembering*), watched videos that illustrated the event (*understanding*), grouped the information focusing on different perspectives of the event (*analyzing*), selected information with higher-quality (*evaluating*) and outlined (*creating*) the structure of the article based on multiple search sessions (p10).

5 DISCUSSION AND CONCLUSION

Our analysis found interesting trends about the information types and cognitive activities involved in the XSS process.

RQ1: What types of information do users need and search for in their everyday cross-session search tasks? Our findings show that XSS often involves *multiple* types of information (i.e., factual/conceptual, procedural, opinion, metacognitive). Sixty-four percent (64%) of our participants looked for at least two of these types of information when working on XSS. Almost every XSS participant’s task (N=106, 96%) required factual/conceptual information. Almost half of the participants required information related to procedural information (43%) or opinions (42%). These results are inline with Byström and Järvelin’s [3] finding that complex tasks often require more types of information than simple tasks.

Different from Byström and Järvelin’s [3] information categorization (i.e., *PI*, *DI*, *PSI*), we adapted a classification schema based on Anderson & Krathwohl’s taxonomy of learning [2]. To the best of our knowledge, our analysis is the first to analyze real-world XSS task complexity through the lens of both the types of knowledge required and the cognitive activities involved. Our results suggest that it can be beneficial to study the knowledge information types involved during real-life XSS based on objective features of the information (e.g., facts, concepts, opinions).

Previous studies found people often need more “information aggregates” and fewer isolated facts for complex tasks [6, 19]. However, our findings show that for complex tasks involving multiple search sessions, participants still searched for a large amount of factual/conceptual information across sessions. This might be because: (1) factual and conceptual information are essential building blocks for understanding and higher-level cognitive activities, (2) participants may have broken down their needs for aggregated information into smaller chunks that could be more easily expressed as keywords for searching, and (3) searchers may need different types of information from various sources to fit for their specific needs.

RQ2: What types of cognitive activities do users perform during cross-session search? We found that XSS involves multiple types of cognitive activities and often requires higher-level cognitive processing. Our analysis showed that a majority of the collected XSS tasks involved 2 to 4 types of cognitive activities (from the six we considered), ranging from finding and understanding the found information to evaluating and utilizing the information to create new knowledge/outcomes. Of the higher-level cognitive activities, *evaluate* was the most commonly noted cognitive activity involved in XSS (65%), followed by *apply* (48%), *analyze* (39%), and *create* (35%). These results are consistent with the experimental finding [24] that single-session complex tasks involve higher-levels of cognitive activities than less complex tasks.

Experimental studies often involve single-session search tasks in which researchers attempt to control or study the specific types of information needed or the types of cognitive activities required for the task (e.g., [6, 8, 9, 24]). Findings show that for simple tasks (e.g., fact-finding tasks) people often rely on lower level cognitive activities [24] whereas for complex tasks people often engage in more higher-level cognitive activities [9, 23]. Our findings suggest that during XSS, complex tasks can be associated with both lower-level and higher-level cognitive activities, depending on the types of information involved. A searcher may look for factual information during initial search sessions and then begin to analyze and evaluate during later sessions as they find more information. Therefore, complex tasks that involve XSS require investigation focusing on both the micro-level of the individual search session and the macro-level treating the overarching task as a whole.

Implications. Our work makes three contributions: (1) It adds to what is known about the characteristics of XSS. We analyzed the complexity of real-life XSS tasks from two dimensions: the involved information types and the types of cognitive activities. Our results suggest that users’ information needs and their cognitive behaviors vary along both dimensions during XSS. A majority of XSS tasks require multiple types of information (or multiple bits of the same type of information). In addition, lower-level and higher-level cognitive activities are coupled through the XSS process. (2) It raises questions for future research about XSS: how users transfer between different types of information and use different cognitive strategies during XSS; What reasons cause these transitions; How search systems can help XSS from the two complexity dimensions? (3) It provide insights for task design and evaluation for studying XSS by experimental studies. Researchers could adapt the XSS task design along the two dimensions to study the effects on users’ XSS behavior.

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